

东南大学 考试卷 (卷)

课程名称 计算结构 (英文) 考试学期 得分

适用专业 信息学院 考试形式 闭卷 考试时间长度 120 分钟

题目	一 (15)	二 (10)	三 (10)	四 (20)	五 (45)	总分
得分						
批阅人						

一、Translate into Chinese (total 15 points)

1. (5 points) The control signals generated by the control unit cause the opening and closing of logic gates, resulting in the transfer of data to and from registers and the operation of the ALU.
2. (5 points) Cache memory is intended to give memory speed approaching that of the fastest memories available, and at the same time provide a large memory size at the price of less expensive types of semiconductor memories.
3. (5 points) A superscalar processor typically fetches multiple instructions at a time and then attempts to find nearby instructions that are independent of one another and can therefore be executed in parallel.

自觉遵守考场纪律 如考试作弊 此答卷无效

姓名

学号

二、 Fill blanks (total 10 points, 2 points/item)

1. There are 136 registers in a computer, 8 of which are left for global use and the others constitute a number of register windows. Each window is consisted of 24 register (8 parameter registers, 8 local registers, and 8 temporary registers). The register window number is _____, and the register window structure supports procedures with depth of _____ at most without window overlap.
2. In designing an instruction pipeline, a variety of approaches can be considered for dealing with conditional branches, such as (), () and branch prediction.
3. An instruction cycle can be divided into several stages, and, at least, each instruction cycle includes two basic stages: () and ().
4. In the three mapping methods of caches, the set associative mapping actually includes the other two techniques. For a cache of 1024 lines, if the set size is () line/lines it is associative mapping, and if the set size is () line/lines it is direct mapping.
5. The infix expression of “**AB+CD-***” is (); The reverse Polish expression of “**A-B-C×D**” is ().

三、 Select A, B, C, or D corresponding to the best answer from the given items (total 10 points, 2 points/item)

1. Computer memory is organized into a hierarchy. The highest level of the hierarchy is ()
A. Cache B. Register C. Memory D. Disk
2. The transfer unit between cache and main memory is ()
A. Bit B. Byte C. 32-bit word D. Block
3. The degree of instruction-level parallelism is determined by the frequency of () and procedure dependency in the code.
A. Read-Write dependency B. Write-Write dependency
C. Write-Read dependency D. Resource conflict
4. There are a number of registers in a processor. The execution time sequence of instructions is controlled by ()
A. MAR B. PC C. MBR D. IR
5. A vertical formatted microinstruction with a 4-bit control field and an 8-bit address field can produce () different control signals at most.
A. 16 B. 12 C. 8 D. 4

四、Answer Questions (total 20 points)

1. **(5 points)** Compared with the use of no pipeline, a two-stage instruction pipeline speeds up instruction execution with the factor of 2 at most, but the two-stage instruction pipeline is unlikely to cut the instruction cycle time in half. Why?
2. **(5 points)** Delayed branch is a technique used in RISC to improve processor performance. What is a delayed branch?
3. **(5 points)** What are characteristics of RISC?
4. **(5 points)** In a super-scalar processor, an instruction window is used. What is the purpose of the instruction window?

五、Problems (total 45 points)

1. **(8 points)** Consider an instruction pipeline of four-stages, fetch instruction (FI), decode instruction (DI), fetch operands (FO), and execute instruction (EI).
 - a. Draw a diagram to show the timing of the instruction pipeline operation for 5 successive instructions without branch;
 - b. Draw a diagram to show the timing of the instruction pipeline operation for the case of a branch is taken in the 4th instruction and jumps to instruction 10;

2. **(10 points)** There are some dependencies in the following code fragment.
 - a. Point out 4 dependencies of which at least,
 - b. Increase instruction parallelism using register renaming.

```
I1: R1 ← R3
I2: R3 ← R3 ADD R5
I3: R4 ← R3 + 1
I4: R3 ← R5 + 1
I5: R6 ← R3 SUB R4
```

3. **(8 points)** A two-way set-associative cache has lines of 16 bytes and a total size of 8Kbytes. The 16Mbytes main memory is byte-addressable.
- Show the address format (tag bits, line/set bits, and word bits) of the main memory addresses using the cache;
 - Give the tag, set, and word offset values of address 6CABBE (in hexadecimal) for the two-way set-associative cache.

4. **(8 points)** Assume a program with six symbolic registers (A, B, C, D, E, F) to be compiled into three physical registers (R1, R2, R3), the active time in the program of each symbolic register is as figure 1. Give a solution of register assignment.

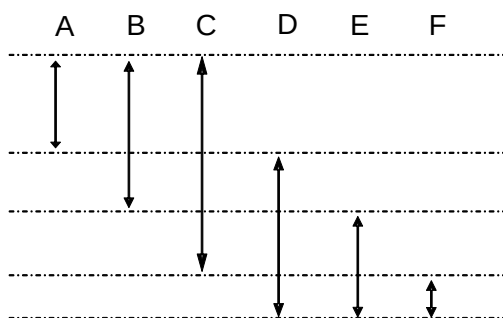


Figure 1. Active time of symbolic registers

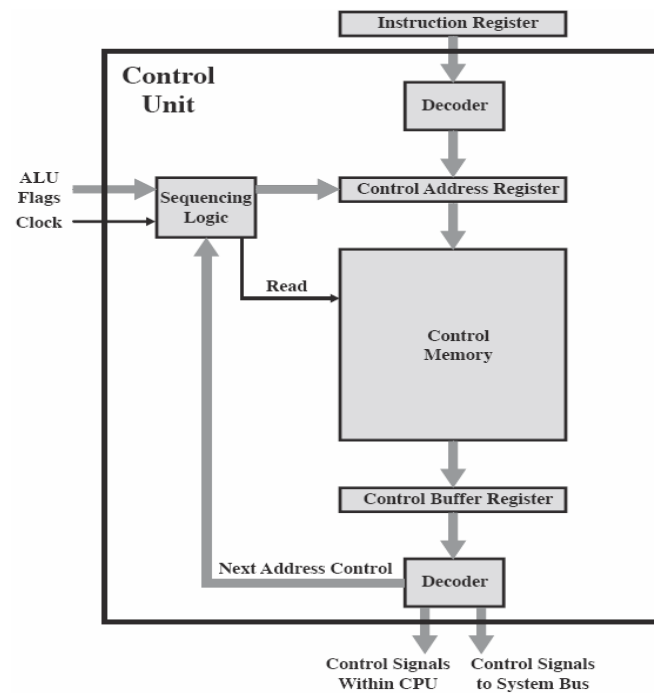


Figure 2. functioning of micro-programmed control unit

5. **(Total 11 Points)** Figure 2 presents the diagram of functioning of micro-programmed control unit. Please try to answer the following questions:

- a. **(2 Points)** What is the purpose of the control memory?
- b. **(2 Points)** What is the difference in function of the upper and the lower **Decoders**?
- c. **(3 Points)** Give the three possible control memory address of the next microinstruction?
- d. **(4 Points)** Describe briefly the work steps of the micro-programmed control unit?